MCQ Question Bank - For Student

Chapter 1: Computer Networks and the Internet

1. Which device forwards packets in the network core?

A) Router B) Monitor C) Printer D) Scanner

2. Transmission rate is commonly measured in?

A) Bits per second B) Bytes per meter C) Packets per user D) Volts

3. Which protocol specifies packet format in the Internet?

A) HTTP B) SMTP C) IP D) DNS

4. Distributed applications mainly run on?

A) Routers B) Switches C) End systems D) DSLAMs

5. The socket interface allows a program to?

A) Repair hardware B) Request data delivery services C) Replace IP D) Encrypt disks

6. A protocol defines message format, order, and?

A) Cable type B) Actions taken on events C) CPU speed D) Screen size

7. DSL commonly uses the existing?

A) Power line B) Telephone line C) Fiber ring only D) Satellite link

8. Cable Internet commonly uses a hybrid?

A) Fiber-coax network B) Radio-only network C) Satellite chain D) Twisted ring

9. FTTH stands for?

A) Fiber To The Home B) Frame To The Host C) Fast Transfer To Hub D) Fiber Through The Highway

10. WiFi is based on which standard family?

A) IEEE 802.11 B) IEEE 488 C) RFC 959 D) OC-3

11. 4G and 5G are examples of?

A) Optical standards B) Wide-area wireless access C) Routing algorithms D) Data centers

12. Physical media that carries signals along a solid medium is called?

A) Unguided media B) Guided media C) Cloud media D) Virtual media

13. Twisted-pair cable consists of?

A) Two insulated copper wires B) Glass tubes C) Satellite links D) Microwaves

14. Which medium is highly resistant to electromagnetic interference?

A) Fiber optics B) UTP only C) Coax only D) Radio

15. Store-and-forward means a router must?

A) Receive the entire packet before forwarding B) Forward first bit immediately C) Ignore headers D) Reserve a circuit

16. Packets waiting for transmission are stored in an output?

A) Browser B) Buffer or queue C) Socket D) Protocol

17. What address is examined by an Internet router for forwarding?

A) MAC only B) Destination IP address C) Email address D) URL text

18. Which approach reserves resources for an entire communication session?

A) Packet switching B) Circuit switching C) Broadcasting D) Caching

19. Packet switching allocates link resources?

A) Permanently B) On demand C) Yearly D) Per user contract only

20. Traffic intensity on a network link is calculated using L, a, and R, where L is the average packet length in bits, a is the average packet arrival rate in packets per second, and R is the link transmission rate in bits per second. Which expression represents the traffic intensity?

A) (L x a)/R B) R/L C) L+R D) a-R

21. Traceroute is used to identify?

A) CPU usage B) Route and delays C) Passwords D) Packet contents only

22. Throughput refers to?

A) Cable length B) Data transferred per second C) Number of routers D) Signal color

23. The bottleneck link is the link with the?

A) Largest delay always B) Smallest transmission rate limiting flow C) Most users D) Newest hardware

24. Layering provides?

A) Modularity B) Infinite bandwidth C) No delays D) Guaranteed security

25. TCP is a protocol of the?

A) Transport layer B) Physical layer C) Link layer D) Application layer

26. UDP provides a?

A) Connection-oriented service B) Connectionless service C) Routing service D) Optical service

27. Packets at the network layer are called?

A) Frames B) Messages C) Datagrams D) Cells

28. Encapsulation occurs when?

A) Headers are added as data moves down layers B) Packets are deleted C) Links are removed D) Queues overflow

29. Malware that spreads from one host to others is?

A) Self-replicating malware B) Passive media C) TDM D) Peering

30. A DDoS attack commonly uses many compromised devices called a?

A) Backbone B) Botnet C) PON D) OLT

Chapter 2: Computer Networks

31. What distinguishes circuit switching from packet switching?

A) Packets always follow different paths B) Resources are reserved in advance C) Links are used only wirelessly D) Data is sent without a route

32. In circuit switching, how long are resources reserved?

A) For one packet only B) Until a queue becomes full C) For the entire communication session D) Only during routing updates

33. Which resource may be reserved by a circuit-switched network?

A) A fraction of link transmission rate B) A destination application C) A DNS name D) A routing protocol

34. Which benefit follows from strict resource reservation after circuit setup?

A) A predictable constant rate B) Unlimited transmission capacity C) No need for switches D) Automatic frequency conversion

35. Which cost is associated with the circuit-switched restaurant analogy?

A) Packet reassembly B) Name translation C) Upfront setup time D) MAC address lookup

36. In packet switching, when are link resources used?

A) On demand B) Only after permanent reservation C) At fixed daily intervals D) Only by Tier-1 ISPs

37. Why may packets wait at a switch in a packet-switched network?

A) A circuit has a guaranteed rate B) The physical layer removes bits C) Arrivals exceed outgoing-link capacity D) An ISP changes its name

38. Packets arrive faster than an outgoing link can transmit them. Which service model explains why delay is possible?

A) Delay-free delivery B) Circuit-guaranteed delivery C) Lossless physical delivery D) Best-effort delivery

39. Why is multiplexing used on circuit-switched links?

A) To remove all protocol headers B) To connect only one host C) To avoid physical media D) To support multiple simultaneous circuits

40. How does FDM separate simultaneous connections?

A) By assigning IP addresses B) By changing packet sizes C) By assigning distinct frequency bands D) By using separate applications

41. A telephone FDM link allocates one traditional band to each call. What bandwidth is assigned per call?

A) 4 Hz B) 4 kHz C) 40 kHz D) 400 MHz

42. In TDM, what is assigned to a connection in every frame?

A) A new random frequency B) The same dedicated time slot C) The entire link continuously D) A different destination port

43. Which statement correctly compares FDM and TDM?

A) FDM separates by frequency; TDM separates by time B) FDM separates by time; TDM by address C) Both reserve only packet buffers D) Both require no resource allocation

44. An access ISP cannot build worldwide links economically. What service should it purchase from a global transit ISP?

A) Application messages B) MAC addresses C) Time slots D) Global reachability

45. Why must competing global transit ISPs interconnect?

A) To reserve one circuit per host B) To replace all regional ISPs C) To preserve universal end-to-end connectivity D) To eliminate routing protocols

46. In the hierarchy shown, which payment path is correct?

A) Tier-1 ISP pays every access ISP B) Regional ISP pays its users C) IXP pays all Tier-1 ISPs D) Access ISP pays regional ISP

47. What is a Point of Presence (PoP)?

A) A transport-layer port B) Provider routers where customers connect C) A reserved TDM slot D) A message reassembly unit

48. Two regional ISPs want traffic to avoid costly upstream transit. Which arrangement should they establish?

A) A circuit between two applications B) A link from a host to itself C) A direct connection between equal-tier ISPs D) A hierarchy with no interconnection

49. What is an Internet Exchange Point (IXP)?

A) A protocol that translates names B) A host that reassembles messages C) A frequency band in FDM D) A facility where multiple ISPs peer

50. Why can peering reduce Internet delivery cost?

A) It increases protocol layers B) Traffic bypasses paid upstream transit C) It reserves more buffers D) It converts frames into messages

51. What characterizes a content-provider network?

A) A network using no protocols B) A single local access link C) A circuit limited to telephone traffic D) A private global network using TCP/IP

52. What is the principal engineering benefit of protocol layering?

A) Infinite bandwidth B) Elimination of headers C) Modularity D) Guaranteed zero delay

53. What service relationship exists between adjacent protocol layers?

A) A layer serves the layer above B) A layer serves only itself C) The top layer serves the physical layer D) Layers never use other layers

54. Which drawback can occur when different layers repeat the same task?

A) Improved isolation B) Simpler addressing C) Duplication of functionality D) Guaranteed throughput

55. What is a cross-layer dependency?

A) A host uses all five layers B) A frame carries a datagram C) A router reads an IP address D) A layer needs information held in another layer

56. A host must transfer e-mail while keeping the task at the application layer. Which protocol should it use?

A) IP B) Ethernet C) SMTP D) DOCSIS

57. What does DNS do in the lecture's example?

A) Translates names to network addresses B) Reserves circuit bandwidth C) Forwards frames by MAC address D) Divides time into slots

58. What is the transport-layer data unit?

A) Bit B) Segment C) Frame D) Message

59. What is the network-layer data unit?

A) Segment B) Frame C) Datagram D) Bit

60. What is IP's role at the network layer?

A) Define datagram fields and router actions B) Translate all domain names C) Reserve frequency bands D) Generate application messages

61. What is the link layer responsible for?

A) Moving data between adjacent nodes B) Routing across the entire Internet C) Generating user content D) Matching application speeds end to end

62. What is the physical-layer data unit?

A) Message B) Segment C) Frame D) Bit

63. Which layers are implemented by a link-layer switch?

A) All five layers B) Application and transport C) Link and physical D) Transport, network, and link

64. At a given layer, how is data from the layer above treated?

A) As a new physical medium B) As a routing provider C) As an empty header D) As an opaque payload

65. What information is included in a transport header in the lecture?

A) Only MAC addresses B) Only frequency bands C) Provider payment records D) Port numbers and error-detection bits

66. Why does the transport layer segment a massive message?

A) To create additional ISPs B) To satisfy MTU limits on network links C) To reserve a circuit in FDM D) To avoid application protocols

Chapter 3: Principles of Network Applications

67. Why are core routers off-limits to application developers?

A) They contain every application B) They use no protocols C) They operate at lower layers D) They cannot forward packets

68. Who designs an application's architecture?

A) The application developer B) The network core C) The access ISP D) The DNS server

69. How do processes on the same host normally communicate?

A) Through global routing B) Through local IPC C) Through HTTP only D) Through an IXP

70. How do processes on different hosts communicate?

A) By exchanging messages B) By sharing CPU registers C) By using one memory space D) By avoiding sockets

71. Which process is the client during a session?

A) The process with more memory B) The process on a server computer C) The process that initiates contact D) The process that closes last

72. In a P2P transfer, a peer that initiates a download has which role?

A) Router B) Server C) Client D) Proxy

73. What is a socket?

A) A router forwarding table B) The API between a process and transport C) A physical network cable D) A Web object

74. Why are two sockets needed for end-to-end process communication?

A) Each packet needs two IP versions B) Each router needs an application C) Each server needs two hostnames D) Each endpoint needs an entry and exit interface

75. What two identifiers are required to address a receiving process?

A) URL and hostname only B) IP address and port number C) MAC address and RTT D) Cookie and status code

76. What does an IP address identify?

A) A process on a host B) A Web object path C) A browser type D) A destination host

77. Why is an IP address alone insufficient for process delivery?

A) IP addresses specify only languages B) One host may run many processes C) Every process uses another host D) IP cannot cross routers

78. Which port is given for an SMTP mail server?

A) 25 B) 21 C) 53 D) 80

79. For destination 128.119.245.12:80, what does 80 identify?

A) The destination host B) The Web-server process C) The HTTP version D) The network path

80. Which application is described as loss-tolerant?

A) File transfer B) E-mail C) Conversational audio or video D) Web documents

81. What characterizes an elastic application?

A) It uses the available throughput B) It requires a fixed minimum rate C) It accepts no variable rate D) It always uses UDP

82. Which application needs tight timing constraints?

A) E-mail retrieval B) File download C) Stored document transfer D) Interactive multiplayer gaming

83. What does endpoint authentication verify?

A) The available bandwidth B) The communicating endpoint's identity C) The URL path length D) The number of objects

84. Which application in the requirements matrix needs no loss and elastic throughput?

A) Internet telephony B) Interactive gaming C) E-mail D) Live video conferencing

85. What does TCP congestion control do during overload?

A) Increases loss intentionally B) Throttles the sender's rate C) Deletes application data D) Disables error checking

86. Which feature distinguishes UDP from TCP?

A) Reliable ordered delivery B) Congestion-controlled sending C) Guaranteed minimum throughput D) No connection handshake

87. Why may a time-sensitive application choose UDP?

A) To guarantee every packet B) To obtain TCP congestion control C) To avoid setup overhead and delay D) To use an HTTP status line

88. Where does TLS operate according to the lecture?

A) In the application layer B) Only in the physical layer C) Inside every router D) In the link layer only

89. Which set of services does TLS add over TCP?

A) Routing, framing, and multiplexing B) Caching, cookies, and URLs C) Encryption, integrity, and authentication D) Ports, paths, and methods

90. What is HTTP?

A) A physical-layer standard B) The Web's application-layer protocol C) A router operating system D) A local IPC mechanism

91. Which two components are essential parts of a URL in the lecture?

A) Hostname and path name B) Port and cookie only C) Method and status code D) RTT and entity body

92. What does a Web browser do after receiving objects?

A) Updates router tables B) Interprets and displays them C) Creates transport protocols D) Assigns server IP addresses

93. Why does HTTP rely on TCP in the lecture's standard model?

A) TCP stores browser cookies B) TCP creates URLs C) TCP displays Web pagesD) TCP provides reliable delivery

94. Which HTTP version in the evolution matrix uses UDP with QUIC?

A) HTTP/1.0 B) HTTP/1.1 C) HTTP/3 D) HTTP/2

95. How many TCP connections are required by non-persistent HTTP for one HTML file and ten JPEGs?

A) Eleven B) One C) Two D) Ten

96. Which delay penalty affects every object in non-persistent HTTP?

A) Half an RTT B) Two RTTs C) Zero RTT D) Ten RTTs

97. Why does non-persistent HTTP create high server overhead?

A) It uses one connection for all objects B) It removes all TCP buffers C) It prevents object retrieval D) It allocates a new TCP connection per object

98. What characterizes a persistent HTTP connection?

A) It closes after every object B) It cannot use TCP C) It remains open for multiple objects D) It sends headers only

99. How many HTTP message types are specified?

A) One B) Two C) Three D) Five

100. What terminates each line in an HTTP/1.1 text message?

A) CRLF B) A cookie ID C) An IP header D) A status code

101. Which HTTP method requests headers without the object body?

A) POST B) PUT C) DELETE D) HEAD

102. Where does POST normally place submitted form data?

A) In the entity body B) In the status phrase C) In the TCP port D) In the hostname